

Batch:	Roll No.:
Experiment / assignment / tutorial No. 07	
Grade: AA / AB / BB / BC / CC / CD / DD	
Signature of the Staff In-charge with date	

Title: Pulse Position Modulation and Demodulation

Objectives:

1. To understand generation process of PPM from PAM and PWM
2. To simulate PPM waveform for a given message signal
3. To reconstruct the original signal through a demodulation process
4. To study the impact of modulation index and pulse width on the signal

OUTCOME:

1. understand working of analog communication transmitter and receiver systems
2. compare Pulse modulation systems and multiplexing techniques.
3. understand noise effect on the performance of communication systems.

- **Theory**

- PPM is a derivative of PWM
- First PWM signal is generated by comparing message signal with a sawtooth carrier. The trailing edges of the PWM pulses change position based on the message.

- **Design Parameters**

- Message frequency (f_m) = 2 Hz
- Carrier/sampling frequency (f_s) = 1000 Hz
- Modulation index: controlled by the amplitude of the message signal
- Pulse width: Defined as a percentage of the sampling period
- Develop a logic for PPM demodulation (Integrator and envelope detector). Embed in the same code and demodulate PPM waveform
- Generate another figure to show PWM, PPM and demodulated signal

- **Design Parameters**

- Open MATLAB Online.
- Develop logic to generate PPM generation and display the message signal, carrier sawtooth signal, PWM signal, PPM signal.
- Observe the PPM plot: notice how the gap between pulses changes as the message signal sine wave goes up and down.
- Change the **pulse_width** variable and observe the changes in the output.

Variations in Parameters

- **Vary f_c (Carrier Frequency):** Increasing f_c increases the number of pulses. This results in a higher resolution and a more accurate reconstruction of the message.
- **Vary Message Amplitude:** If the message amplitude exceeds the carrier amplitude, over-modulation occurs, and pulses will be lost or merged.

Somaiya Vidyavihar University
K J Somaiya School of Engineering

- **Post lab activity:**

- Shift in the position of each pulse of PPM depends on what?
- State practical applications of PPM.

Signature of faculty in-charge